Unleashing UVM™

PACKET SPLITTER

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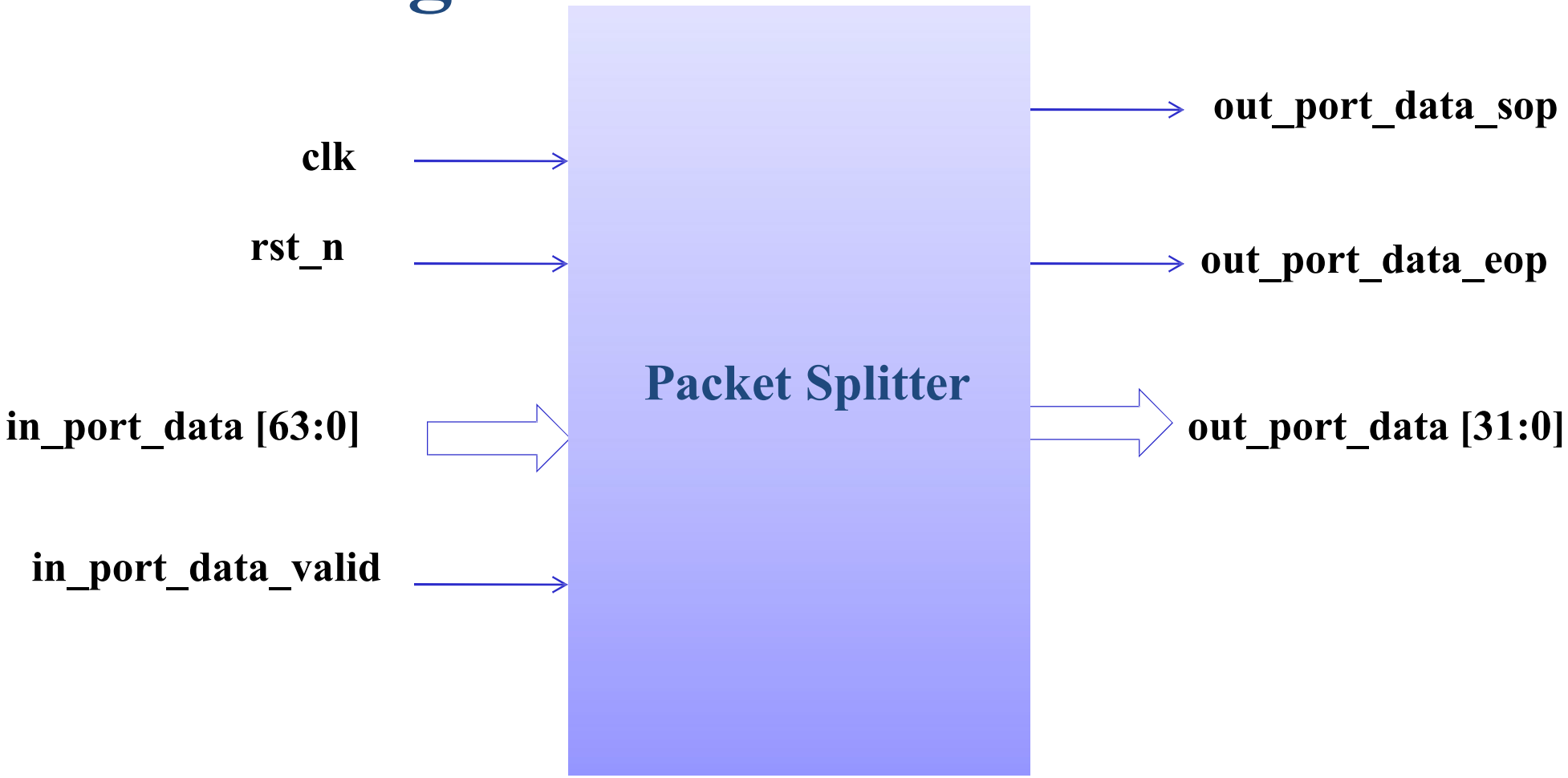
Block Diagram:



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Signal Description:

Signal	Width	Direction	Description
clk	1 bit	Input	Clock input
rst_n	1 bit	Input	Active low reset
in_port_data	64 bits	Input	Data input
in_port_data_valid	1 bit	Input	Validates input data
out_port_data	32 bits	Output	Data output
out_port_data_sop	1 bit	Output	Indicates start of packet for output data
out_port_data_eop	1 bit	Output	Indicates end of packet for output data

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Description:

Input Port :

- ❑ "in_port_data_valid" validates the data transfer .
- ❑ "in_port _data" specifies the input data to be transferred.

Output port:

- ❑ "out_port_data" fetch the data from the input port when Valid signal is high.
- ❑ "out_port_data_sop" is a pulse indicate the start of data packet transfer.
- ❑ "out_port_data_eop" is a pulse indicate the end of data packet transfer.

Note :

- ❑Variable packet size to be sent.
- ❑Design supports Big-endianness
- ❑Till one data transfer gets completed, another new transfer should not take place.

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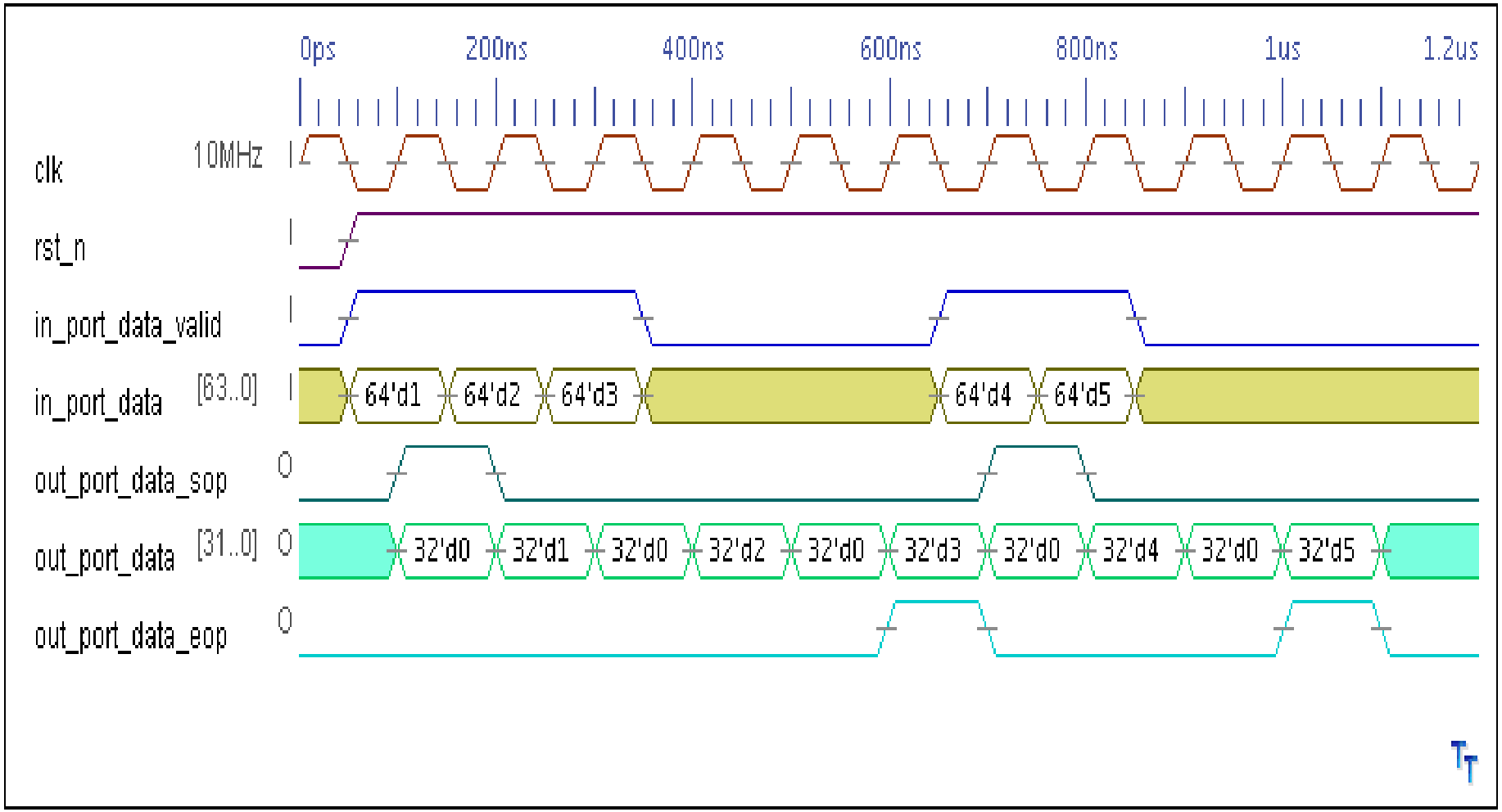
Timing_Diagram



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